

RASHMI VENKANGOUDA PATIL

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SUMMARY

Software Development Engineer with 3+ years of experience building production backend systems and AI-powered applications. Proficient in designing agentic AI workflows, RAG pipelines and LLM integrations using LangGraph, LangChain and Claude AI. Experienced in Python, FastAPI, React.js and cloud-native microservices on AWS with Docker and Kubernetes.

EDUCATION

Arizona State University, Tempe, Arizona, USA | GPA – 4.0/4.0 **January 2024 - December 2025**

Masters in Information Technology: AI, Natural language processing, Big Data, Machine Learning, AWS

Basaveshwar Engineering College, Bagalkot, Karnataka, India | CGPA - 7.75/10 **August 2016 - August 2020**

Bachelors in Computer Science and Engineering: System design, Data Structures, Software Engineering, DBMS

TECHNICAL SKILLS

- **AI/ML and GenAI:** LangGraph, LangChain, RAG, Prompt Engineering, Vector Databases, LLMs, Langfuse, CrewAI
- **Programming Languages:** Python, JavaScript, TypeScript, Java, SQL, C#, Go
- **Frontend Development:** React.js, Angular, HTML5, CSS3
- **Backend and Database:** FastAPI, Django, Node.js, PostgreSQL, MongoDB, MySQL
- **Cloud and DevOps:** AWS, Docker, Jenkins, GitHub Actions, Databricks

WORK EXPERIENCE

Software Development Engineer | Kingship AI Inc, San Jose, California **February 2026 - Present**

(Python, FastAPI, React.js, Google Gemini 2.5, PyMuPDF, MongoDB, Pydantic)

- Engineered **GenAI pipelines** using **Gemini 2.5 Pro** and **PyMuPDF** for intelligent **PDF** and **DOCX** document processing
- Designed **structured prompt engineering** across **3 Gemini 2.5 Flash** backend services with **Pydantic**-validated **JSON** outputs
- Developed multi-step **React.js** wizard with **FastAPI** integration feeding structured user inputs to **Gemini API**
- Built 10+ **FastAPI** endpoints with **MongoDB** covering full **CRUD**, **LLM** integration and production workflows

Software Engineer | Cognizant Technology Solutions, Chennai, India **July 2021 - December 2023**

(React.js, Node.js, Python, Django, RESTful APIs, Microservices, Docker, Kubernetes, AWS, Jenkins, SQL, Git/GitHub)

- Built **RESTful APIs** and **microservices** using **Node.js** and **Django** across 5+ production applications
- Developed reusable **React.js** components integrating frontend with backend APIs across 3+ client applications
- Implemented **JWT** and **OAuth** authentication securing microservices with **role-based access control**
- Optimized **SQL** queries and schema design reducing average backend response time by **30%**
- Containerized **microservices** with **Docker**, built **Jenkins CI/CD pipelines** and orchestrated **Kubernetes** deployments on **AWS**
- Resolved **200+ production incidents** across backend services maintaining **SLA compliance** and reducing downtime

AI Engineer Intern | Rubix Pvt Ltd, Bangalore, India **January 2021 - June 2021**

(Python, Scikit-learn, TensorFlow, Matplotlib, Keras, ML algorithms, Tableau)

- Built and tuned **ML models** including **Random Forest**, **KNN** and **Decision Trees** improving precision by **12%**
- Performed **EDA** on large datasets using **Pandas**, **NumPy** and **Seaborn** extracting actionable business insights
- Presented analytical findings via **Tableau** dashboards to non-technical stakeholders

PROJECTS

Multi-Agent SOC Automation System *Tech: Python, FastAPI, LangGraph, Claude AI, Qdrant, PostgreSQL*

- Built **8-agent LangGraph** pipeline with **supervisor routing**, **conditional edges** and **human-in-the-loop** approval
- Implemented **hybrid RAG** retrieval using **Qdrant** and **BM25** with **Cohere reranking** for alert triage
- Evaluated pipeline across 20 test cases using **RAGAS** tracking **faithfulness** and **context recall** scores

Backend Performance Monitor *Tech: Python, FastAPI, PostgreSQL, Prometheus, Grafana, Docker, Locust*

- Built **7-service observability** platform monitoring real-time **API metrics** via **Prometheus** and **Grafana**
- Engineered **JWT**-secured microservice authentication with automated **threshold-based alerting** pipeline
- Load tested system using **Locust** simulating **50 concurrent users** validating performance under peak load

Adaptive RAG System for Technical Documentation *Tech: Python, FastAPI, LangChain, Claude AI, Qdrant, Celery*

- Built multi-source ingestion pipeline using **PyMuPDF**, **GitPython** and **BeautifulSoup4** with **SemanticChunker** and **CodeSplitter**
- Implemented **HyDE** query expansion and **MultiQueryRetriever** with **Celery** async task queue for concurrent document ingestion
- Designed **feedback-aware reranking system** shipping user ratings to **Snowflake** pipeline with **Langfuse** observability tracing